

Literature Review: Privacy-Preserving e-KYC and Decentralized Identity

Practice School-I (PS-1) Technical Log - Day 1
CDAC Pune

Topic: Core Zero-Knowledge Cryptography & Identity Verification

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Abstract

This document synthesizes four foundational academic papers regarding the application of Zero-Knowledge Proofs (ZKPs) to electronic Know Your Customer (e-KYC) and decentralized identity (DID) frameworks. The review categorizes the current research into selective disclosure mechanics, non-transferable reputation systems, stateless authentication, and off-chain zkVM attestations. This literature review serves as the architectural baseline for the development of a privacy-preserving Indian e-KYC prototype.

1 Selective Disclosure via Merkle Trees and zk-SNARKs

Core Paper: *A Privacy-Preserving Selectively Disclosed eKYC System Using Merkle Tree and Zero-Knowledge Proofs* (Ahmed et al.)

Traditional centralized databases present severe vulnerabilities to data breaches and unauthorized surveillance. To counter this, Ahmed et al. propose a system that enforces data minimization through "selective disclosure".

- **Merkle Tree Integration:** Instead of storing raw identity data, the system structures user attributes (e.g., Name, Date of Birth) as leaf nodes in a Merkle Tree, obfuscated using a Poseidon hash function alongside cryptographic salts.
- **Single Proof Generation:** The issuer digitally signs the Merkle Root and commits it to the blockchain. The user can then generate a single Groth16 zk-SNARK proof demonstrating possession of specific attributes (like $\text{Age} \geq 18$) without revealing the entire credential document.
- **Performance Context:** The paper benchmarks proof generation at approximately 312 milliseconds and on-chain verification at 20,000 gas, proving its viability for scalable financial applications.

2 Decentralized Reputation and Soulbound Tokens (SBTs)

Core Paper: *ZKP Enabled Identity and Reputation Verification in P2P Marketplaces* (Kalbantner et al.)

In decentralized and Peer-to-Peer (P2P) financial marketplaces, maintaining regulatory compliance (AML/CTF) while preserving transactional privacy is a primary challenge.

- **Identity Anchoring:** The authors integrate Verifiable Credentials (VCs) with Soulbound Tokens (SBTs). SBTs are non-transferable NFTs that securely bind a user's verified KYC status directly to their Decentralized Identifier (DID).
- **Tiered Reputation System:** To counter Sybil attacks and incentivize good behavior, the framework introduces Marketplace Reputation Tokens (MRTs). These tokens establish a 3-tier reputation scoring mechanism based on transaction history and varying depths of KYC verification.
- **State Channels:** Trades are initialized through off-chain state channels where ZKPs are used to verify the SBTs and MRTs without publicly exposing the users' actual identities on the ledger.

3 Stateless Authentication via Human-Readable UIDs

Core Paper: *ZKPass - A Zero-Knowledge Proof Based Authentication System for Secure Identity Verification* (Srinivas P M et al.)

A critical barrier to mainstream Web3 adoption is the complexity of cryptographic key management and the vulnerability of password-based logins. ZKPass proposes a lightweight, passwordless alternative.

- **Ephemeral Key Generation:** The framework derives temporary private keys locally from a Unique Identifier (UID) exclusively at the moment of login. This eliminates the need to store long-term keys or passwords, drastically reducing the attack surface.
- **Self-Sovereign Recovery:** Moving away from centralized recovery agents, the system generates a human-readable English recovery phrase from the UID and a secret salt during registration.
- **Storage Efficiency:** By storing only the hash of the UID on the blockchain rather than complete credential proofs or public keys, the system mitigates blockchain congestion and ensures strict identity privacy.

4 Off-Chain Attestations via EAS and zkVM

Core Paper: *Zero-Knowledge Identity Verification* (Kumar et al.)

Publishing attestations directly on-chain inherently exposes relational metadata, violating privacy-by-default principles. Kumar et al. decouple the attestation issuance from on-chain verification.

- **EIP-712 Standard:** The system relies on the Ethereum Attestation Service (EAS) to issue off-chain, structured attestations signed securely under the EIP-712 standard.
- **zkVM Processing:** Instead of writing bespoke arithmetic circuits, the attestations and their logical constraints are processed inside a Zero-Knowledge Virtual Machine (specifically SP1).
- **Client-Side Verification:** The zkVM compiles succinct Groth16 or Plonk proofs that are lightweight enough to be verified completely client-side using WebAssembly (WASM) in a standard web browser or Node.js environment.

5 Synthesis for PS-1 Project Implementation

The literature confirms that standard on-chain KYC systems are structurally inadequate due to data exposure and gas inefficiencies. To build the required India-centric e-KYC prototype for this Practice School project, the architecture will synthesize three core mechanisms from the reviewed literature:

1. **The Data Structure:** We will adopt the *Merkle Tree* approach (Ahmed et al.) to hash the individual fields of a DigiLocker JSON credential, allowing for selective disclosure (e.g., Age).
2. **The Proof Generation:** To ensure broad compatibility and reduced manual circuit errors, we will explore generating the proofs via a *zkVM environment* (Kumar et al.) rather than raw Circom compilation.
3. **The Storage Layer:** To minimize blockchain footprint, we will adopt the *Stateless UID protocol* (Srinivas et al.), ensuring the smart contract verifies the zk-SNARK proof without storing any persistent state beyond a single cryptographic hash.